

Leveraging Simulated Sensor Data and Foundation Models for Human Motion Understanding

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Abstract

Sensor-based human motion understanding is a fundamental task in ubiquitous computing, enabling applications in health monitoring, pose estimation, and activity recognition. However, real-world data collection is often costly, intrusive, and difficult to scale in various conditions. This doctoral research introduces a framework that integrates sensor simulation, multimodal data fusion, and foundation models to create robust, generalizable, and semantically rich human motion understanding (HMU) systems. This work builds on several original contributions, including the generation of simulation-based sensor data, the estimation of simulation-embedded poses, and the training of sensor foundation models using synthetic data. These efforts demonstrate how simulated data can substitute for or augment real-world data in HMU tasks, while leveraging large pretrained models and contrastive learning for motion-to-text alignment and semantic generalization. Preliminary results confirm strong sim-to-real transfer, improved cross-sensor performance, and improved interpretability in motion understanding. The ultimate goal of this research is to develop a low-cost, scalable pipeline for motion interpretation that minimizes the reliance on extensive physical data collection and enhances adaptability across environments, users, and sensing modalities.

CCS Concepts

• Computing methodologies → Learning latent representations; Modeling and simulation.

Keywords

HAR, pose estimation, HMU, sensor simulation, pretraining, foundational model, sensor

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1 Problem Statement

Human Motion Understanding (HMU) [2] is a broad and foundational area of ubiquitous computing that involves sensing, modeling, and interpreting human movement from raw sensor data. Its scope spans pose estimation [14], action and gesture recognition [11], gait analysis [12], and motion-driven applications such as virtual garment simulation [5], music-based music based dance choreography generation [29], and human-object interaction [34].

Despite decades of research, building robust, generalizable, and semantically meaningful HMU systems remains a great challenge. Current models depend heavily on large amounts of high-quality real-world sensor data [13], often collected under controlled conditions using specialized motion capture setups or carefully placed wearable devices. Collecting such data at scale is time-consuming, costly, and intrusive for participants. Moreover, models trained on controlled datasets often fail to generalize to new users, sensor placements, clothing variations, or complex real-world environments [36].

While recent advances in foundation models and large pretrained architectures like ImageBind [9], IMU2Clip [15] etc have shown promise in computer vision and natural language processing, the development of domain-specific foundation models for human motion understanding is still in its infancy. Off-the-shelf foundation models are not directly suitable for sensor-based motion data, and there is a lack of large, diverse, realistic motion datasets required to train them effectively.

To address these limitations, this research proposes a novel approach that starts by designing realistic, physics-based sensor simulation pipelines from scratch. These custom-built simulators generate synthetic multimodal sensor data—including inertial, pressure, and visual streams—under controlled yet diverse conditions, emulating real-world variations in human movement, sensor placement, and environment. This rich simulated data is then used to train and adapt large foundational models specifically for HMU, enabling them to learn generalizable, semantically rich representations that bridge the gap between synthetic and real data.

Therefore, the central problem addressed in this doctoral research is: How to design and utilize advanced sensor simulation techniques in combination with large foundational model architectures to develop robust, domain-specific foundation models for human motion understanding that minimize dependence on costly real-world data and generalize effectively across diverse contexts, sensors, and applications.

2 Related Work

Physics-Based Simulation Beyond Human Motion Understanding. Physics-based simulation has become a powerful tool in various

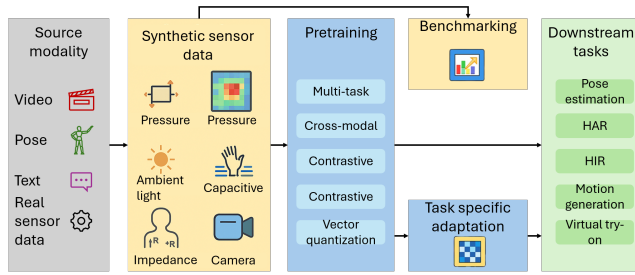


Figure 1: The proposed end-to-end pipeline for sensor-based human motion understanding: (1) realistic physics-based and hybrid simulation generates diverse multimodal signals; (2) multi-stream foundation models learns unified latent representations via multi-task, cross-modal, and contrastive learning; (3) the pretrained model generalizes to real data and adapts with minimal supervision.

fields to address real-world problems that are otherwise costly, unsafe, or impractical to study exclusively through physical experimentation [30]. In robotics, simulators enable training of control policies and reinforcement learning agents in virtual environments before deploying them on physical robots, reducing wear-and-tear and safety risks [27, 31]. In autonomous driving, large-scale simulated environments allow the development and testing of perception and planning systems under diverse traffic scenarios, weather conditions, and rare edge cases [7, 26]. In computer graphics and virtual reality, physics-based simulators produce realistic animations, cloth dynamics, and soft-body deformations for movies, gaming, and digital twins [1]. Across domains, simulation offers a safe, controllable, and reproducible framework for generating diverse data and testing under varied conditions without incurring real-world costs.

Rising Importance of Sensor-Based Human Motion Understanding. Sensor-based human motion understanding (HMU) is becoming increasingly relevant due to the widespread adoption of wearable devices, ambient sensors, and smart textiles [4]. Applications include remote health monitoring, rehabilitation, activity recognition, context-aware human-computer interaction, and immersive virtual experiences [32]. Compared to vision-only systems, wearable and embedded sensors provide continuous, privacy-preserving, and robust tracking of human movement, functioning reliably even under poor lighting or occlusion [35]. However, developing robust HMU models remains challenging due to their dependence on large, diverse, and high-quality sensor datasets that capture variations in user behavior, sensor placement, clothing, and environmental factors [10]. Collecting such data is costly, time-intensive, and often intrusive, which limits scalability and diversity.

Limitations of Task-Agnostic Foundation Models for Sensor Data. Large foundation models have achieved remarkable success in natural language processing and computer vision by learning task-agnostic representations from massive, heterogeneous datasets [3, 17]. These models capture high-level semantics and can adapt to various downstream tasks with minimal fine-tuning. However, applying such models directly to raw sensor data—such as inertial signals, pressure readings, or ambient light measurements—poses

unique challenges [33]. Sensor data differ significantly from text or images in terms of temporal dynamics, noise characteristics, and sampling irregularities. Furthermore, general-purpose pre-trained models lack exposure to the specific physical patterns and constraints inherent in human motion signals, which limits their ability to interpret fine-grained movement or generalize across sensing configurations. Consequently, there remains a gap in developing domain-specialized foundation models tailored for sensor-based HMU.

Current Approaches and This Research Direction. Recent studies have explored physics-informed neural networks, domain adaptation, and synthetic data generation to improve the robustness of sensor-based HMU [18]. Some approaches use biomechanics engines or motion capture datasets to synthesize pose trajectories, while others generate synthetic inertial or pressure data for specific recognition tasks [6]. However, these pipelines often treat simulation and learning as independent stages and typically focus on task-specific models rather than developing a general-purpose, reusable HMU foundation model. Additionally, existing simulators are often generic and do not faithfully replicate sensor noise, placement variability, or complex human-environment interactions [16].

This research addresses these gaps by pursuing two key directions: (1) designing realistic, physics-based sensor simulation pipelines from scratch, modeling wearable and ambient sensor signals under diverse and controllable conditions; and (2) integrating this synthetic data with large foundation model architectures to train domain-specific representations that bridge the gap between synthetic and real sensor data. This unified approach aims to enable robust, scalable, and semantically rich human motion understanding across a wide range of tasks and real-world applications.

3 Methodology

This work proposes a unified framework that combines flexible sensor signal generation with large-scale representation learning to build a robust, domain-specific foundation model for human motion understanding (HMU). The methodology is designed to handle diverse sensor modalities by mixing physics-based simulation, deep learning-based synthesis, hybrid approaches, and physics-informed neural networks (PINNs). It also integrates external knowledge sources such as video pose data and text embeddings to strengthen semantic alignment.

3.1 Sensor Signal Generation

The framework leverages multiple data sources to cover a wide range of real-world variability:

- **Video and Pose Data:** Public motion capture datasets and vision-based pose estimators provide kinematic trajectories used as priors to drive simulators or train signal generators.
- **Textual Annotations:** Natural language descriptions and activity captions enable semantic supervision through contrastive learning and open-vocabulary alignment.
- **Stable Sensor Modalities:** When available, high-quality sensor channels (e.g., reference IMU data) help calibrate simulators and guide the synthesis of more volatile or hard-to-measure signals like pressure or ambient light.

Depending on the target sensor modality and available resources, the pipeline dynamically selects among four strategies:

- **Pure Deep Learning Generation:** For well-understood signals with abundant labels (e.g., IMU conditioned on pose), sequence-to-sequence models learn to map pose trajectories to synthetic sensor readings.
- **Hybrid Physics + Supervised Learning:** For complex signals like garment-induced pressure maps, coarse physics-based simulation provides initial estimates refined by supervised learning models trained on real measurements.
- **Physics-Informed Neural Networks (PINNs):** For signals where explicit physics constraints must be respected (e.g., force balance, contact pressure), PINNs combine data-driven learning with embedded physical laws.
- **Pure Physics Simulation:** For signals directly determined by biomechanical dynamics (e.g., contact forces, joint torques), custom physics engines produce high-fidelity ground truth.

This modular approach ensures realistic, diverse, and controllable synthetic data even for rare conditions and edge cases.

3.2 Foundation Model Architecture and Pre-training Strategies

The core learning component is a multimodal foundation model tailored to sequential sensor data. Its architecture uses a multi-stream transformer encoder: each modality (IMU, pressure, ambient signals) has a dedicated encoder; outputs are fused via cross-attention to learn a unified latent representation.

To train robust and semantically rich embeddings, multiple strategies are combined:

- **Multi-Task Learning:** The model is trained to perform related tasks such as pose estimation, activity classification, and contact prediction concurrently, encouraging transferable features.
- **Cross-Modal Alignment:** Reconstruction losses align signals across modalities (e.g., predicting pressure maps from IMU streams) to enforce consistency and redundancy.
- **Contrastive Learning:** Inspired by vision-language models, a contrastive objective aligns sensor embeddings with textual motion descriptions or sentence embeddings from large pre-trained language models, enabling open-vocabulary querying.
- **Vector Quantization (VQ):** A VQ layer discretizes latent codes, promoting compact and structured representations, which supports more interpretable downstream tasks.

This training pipeline uses large-scale synthetic data and limited real recordings to achieve strong generalization.

3.3 Domain Adaptation and Deployment

Once pre-training is complete, the foundation model is evaluated directly on real-world data in a zero-shot setting to test its generalization capability. In cases where the simulated signals are sufficiently realistic and the variability is well-covered, the model can perform robustly without further fine-tuning. For scenarios with domain gaps or application-specific nuances, lightweight domain adaptation or supervised fine-tuning can be applied using a small

amount of real sensor data. This flexible deployment strategy allows practitioners to balance training cost and performance, supporting both plug-and-play zero-shot inference and tailored adaptation for critical tasks.

This methodology integrates flexible, physics-aware simulation, data-driven signal generation, and advanced multi-task, cross-modal learning to produce a robust, semantically grounded foundation model for sensor-based human motion understanding. It directly addresses the challenges of data scarcity, domain shift, and semantic interpretability, paving the way for scalable and practical HMU systems.

4 Evaluation

The proposed framework will be rigorously evaluated in three key dimensions: (1) the realism and coverage of the synthetic sensor data, (2) the zero-shot generalization capability of the domain-specific foundation model, and (3) the effectiveness of lightweight domain-specific adaptation where needed. Each dimension includes detailed metrics and test conditions to validate the design choices.

4.1 Synthetic Sensor Data Quality

To ensure that the physics-based and hybrid-generated signals are realistic and diverse, multiple quantitative and qualitative measures will be used:

- **Statistical Similarity:** Compute Pearson correlation, root mean square error (RMSE), and Dynamic Time Warping (DTW) distance between simulated signals and real sensor recordings for the same motions.
- **Distribution Alignment:** Use Kullback-Leibler (KL) divergence and Earth Mover's Distance (EMD) to compare the probability distributions of simulated vs. real signal magnitudes and frequency spectra.
- **Variability Coverage:** Perform parametric sweeps in the simulator (e.g., varying joint angles, cloth stiffness, sensor placement offsets) and show that the generated signal space spans the natural variation observed in real datasets.
- **Utility for Learning:** Train baseline HAR and pose estimation models purely on synthetic data and report their absolute and relative performance when tested on real recordings — demonstrating practical value beyond raw signal similarity.

4.2 Foundation Model Generalization

The core goal is to verify that the foundation model trained on synthetic data performs robustly on real-world tasks without explicit domain tuning. This will be validated by:

- **Zero-Shot Accuracy:** Report standard classification metrics (accuracy, F1-score) for activity recognition on public HAR datasets using only the pre-trained model.
- **Pose Estimation Error:** For tasks involving pose recovery, report mean per joint position error (MPJPE) in centimeters compared to motion capture ground truth.
- **Contact & Prediction:** Evaluate mean absolute error (MAE) and normalized root mean square error (NRMSE) for sensor localization predictions.

- **Robustness Tests:** Introduce controlled perturbations e.g., simulate sensor misplacement, different walking speeds, or novel activities — and measure performance degradation to assess domain coverage.
- **Cross-Modal Semantics:** For open-vocabulary tasks, use top-1 and top-5 retrieval accuracy for motion-to-text and text-to-motion matching, leveraging pre-trained language embeddings as ground truth anchors.

4.3 Domain-Specific Adaptation

In practical settings where a domain gap persists, the benefit of lightweight adaptation will be tested by:

- **Few-Shot Fine-Tuning:** Fine-tune the foundation model on small labeled subsets (e.g., 1%, 5%, 10% of the real dataset) and report accuracy gains relative to the zero-shot baseline.
- **Data Efficiency Curves:** Plot performance vs. the number of fine-tuning samples to demonstrate how quickly the model adapts, proving cost-effectiveness compared to training from scratch.
- **Task Stability:** Measure whether adaptation on one task (e.g., new activity class) preserves generalization on unrelated tasks (e.g., pose estimation), highlighting representation reuse.

4.4 Rationale

This evaluation design directly tests the central hypotheses: (i) high-fidelity synthetic data can approximate real sensor signals well enough for training, (ii) the foundation model learns robust, transferable representations capable of zero-shot inference, and (iii) when needed, minimal real data suffices to close remaining gaps efficiently. Together, these metrics demonstrate technical soundness and practical scalability for real-world ubiquitous computing applications.

5 Contributions and Future Work

This doctoral research aims to advance sensor-based human motion understanding (HMu) by combining custom simulation, domain-specific foundation models, and scalable deployment strategies. The contributions are organized into completed work to date and the planned next steps.

5.1 Progress To Date

- **Custom Sensor Simulation Pipeline:** A first version of a physics-based simulation pipeline has been developed, capable of generating realistic inertial [8, 23, 28, 37], capacitive [19], bio-impedance [21] and pressure signals [22, 25] driven by motion capture data and biomechanical models. Initial experiments demonstrate controllable parameter variation for human morphology and clothing properties [5].
- **Domain-Specific HMu Foundation Models:** Early versions of a multimodal transformer-based foundation model have been pre-trained on synthetic IMU [8] and pressure data [20, 24]. Contrastive pre-training and Vector quantization has been tested for aligning sensor embeddings with pose sequences and basic activity labels.

- **HMu Inference:** Preliminary results show that models trained purely on simulated data achieve promising zero-shot and few-shot performance on public HAR datasets without task-specific fine-tuning, confirming the feasibility of sim-to-real transfer for common activities.
- **Comparative Experiments:** Initial ablations comparing pure physics simulation and hybrid simulation-plus-learning pipelines for pressure data generation have been conducted, providing insights into fidelity vs. computational cost trade-offs.

5.2 Planned Contributions and Future Work

Building on this foundation, the research will extend in the following directions:

- **Refinement of the Simulation Pipeline:** Develop a more modular and high-fidelity simulator that supports additional modalities (e.g., bio-impedance, EMG), with improved physics realism and parameterized noise models.
- **Advanced Foundation Model Training:** Expand the foundation model to integrate multi-task learning, robust cross-modal reconstruction, contrastive objectives with external sentence encoders, and vector quantization for compact representations.
- **Adaptable HMu Pipeline:** Systematically validate zero-shot generalization across broader scenarios, and develop lightweight domain adaptation modules to handle residual domain gaps with minimal labeled data.
- **Detailed Comparative Analysis:** Conduct a large-scale, systematic comparison of signal generation strategies (pure physics, hybrid, PINN, deep learning-only) and pre-training objectives, quantifying their impact on generalization, zero-shot accuracy, and adaptation efficiency.
- **Real-World Deployment:** Integrate the final foundation models into practical ubiquitous computing prototypes, such as smart wearables for continuous health monitoring and motion-aware ambient systems.

Together, this staged work aims to overcome key data scarcity and domain shift barriers in HMu, paving the way for scalable, semantically rich, and practically deployable motion understanding in everyday ubiquitous computing contexts.

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